

Mohamed Abdelrahman

832-419-9579 | msabdelrahman02@gmail.com | <https://www.linkedin.com/in/msabdelrahman02> |
<https://github.com/mohameds-dev> / mohamed-a-portfolio.vercel.app

EDUCATION

University of Houston

Bachelor of Science in Computer Science

- GPA: 3.28 / 4.00

Aug. 2023 – May 2026

Houston, TX

EXPERIENCE

Code[Coogs] Competitions Director

Jan 2025 – May 2025

University Of Houston

Houston, TX

- Led 9 competitive programming events on data structures, algorithms, and complexity analysis to 30+ students
- Organized algorithmic contests on Kattis, curating problem sets aligned with LeetCode-style interview topics
- Hosted interactive workshops with live coding and peer discussions to improve student engagement
- Collaborated with student leaders and faculty to expand outreach and iterate on event formats

Coding Expert - AI Trainer (Contract)

Aug 2023 – Jan 2025

Outlier.ai

Remote

- Developed solutions for fine-tuning code generation models with Python and C++, improving quality by 40%
- Created prompts to produce model deviations that improve model training through prompt engineering techniques
- Annotated LLM output for over 400 responses and rated its quality based on a rigorous rubric
- Reviewed submissions for other contributors and provided feedback, improving task quality by 30%

Software Engineering Intern

Aug. 2022 – Jan 2023

bld.ai

Remote

- Collaborated with four interns to develop a full-stack application with React, Django, and PostgreSQL
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month, following a two-week sprint cycle
- Adhered to Agile Practices with sprint planning, daily stand-ups, and retrospectives to optimize workflow by 50%

PROJECTS

TallyCart (In Progress) | *Django, EasyOCR, Prompt Engineering*

- Designing a Django backend to structure purchase data and product price history for optimized shopping recommendations
- Building relational data models and logic to parse and organize large volumes of receipt-derived data
- Integrating EasyOCR with LLM-powered postprocessing to convert raw receipt text into structured product records

C++ Ordered Map Library | *C++17, CMake, LCOV, Valgrind, Catch2, GitHub Actions, TDD*

- Implemented Ordered Map data structure by building a doubly linked list from scratch with smart pointers and using an unordered map for O(1) lookups and insertions while maintaining insertion order
- Achieved 97.3% code coverage across 2,246 test assertions with Catch2 for comprehensive testing
- Designed STL-compatible interface with iterator support and exception safety through modern C++17 features
- Established CI/CD pipeline with automated testing through GitHub Actions and CMake
- Verified zero memory leaks through Valgrind analysis and created comprehensive documentation

Nota | *Flutter, Dart, Bloc, SQFlite, Quill, Material 3*

- Developed rich-text note-taking app for Android with offline storage through SQL queries in SQFlite
- Integrated Flutter Quill editor to support formatting such as bold, italics, headers, links, and lists
- Implemented labeling and filtering features for efficient note organization and retrieval
- Managed UI and business logic with Bloc state management for clean separation of concerns
- Designed responsive UI with Material 3 components and support for light/dark theme switching

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart

Frameworks: Django, React, NextJS, Flutter

Developer Tools: Git, Docker, Postman, VS Code, CLion, Visual Studio, PyCharm, IntelliJ, DBeaver

Libraries: unittest, React Router, Bloc, Flutter Quill, SQFlite, Axios